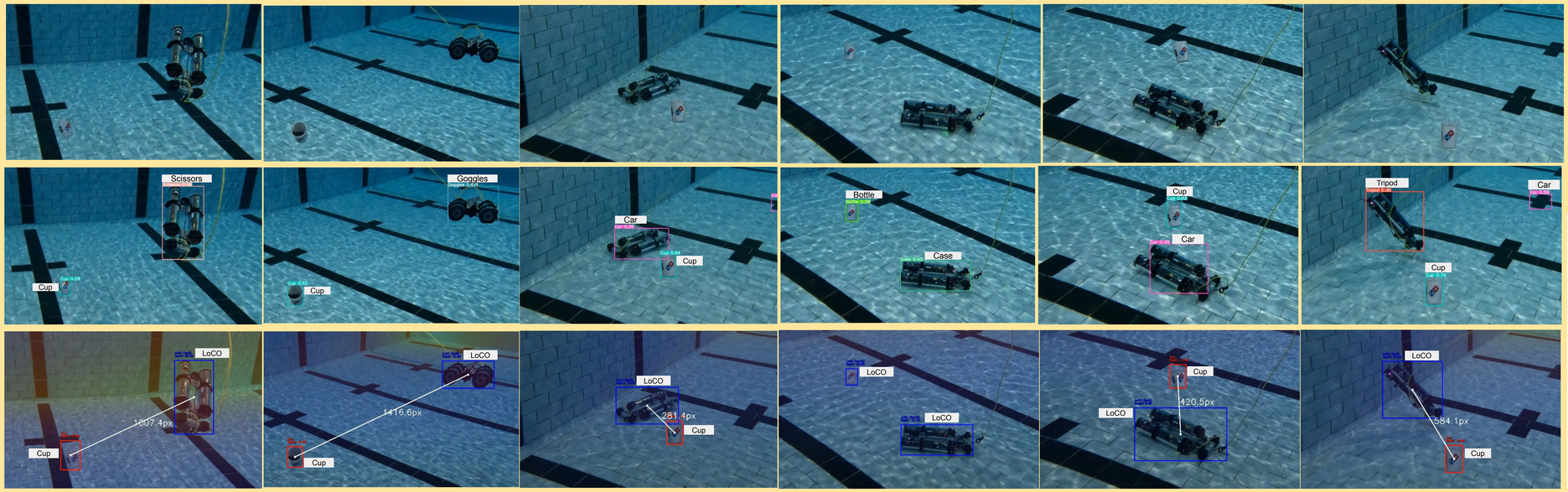
Overcoming Data Scarcity in Child AUV and Target Position Estimation for Underwater Trash Removal Using SIFT-Enhanced YOLOv9

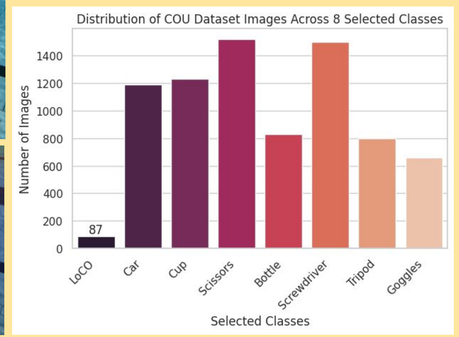
Francis Aweenagua, Amreen Hossain, Raian Haider Chowdhury, Nour Hany, Junaed Sattar

Computer Science & Engineering · Minnesota Robotics Institute, University of Minnesota · IRVLAB · BioSENSING & BioROBOTICS Laboratory



Frame-wise comparison of YOLOv9 Only and YOLOv9 + SIFT position estimation. YOLOv9 misclassified the AUV while SIFT corrects the missed classifications, showing improved performance.

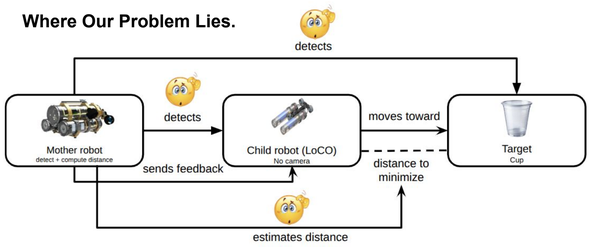
COU Dataset



Motivation

- A single robot is less effective in underwater trash removal.
- Multiple AUVs are financially demanding.
- Some AUVs might face form factor and functionality limitations in removing trash.
- We can use multiple custom-made and resource constrained AUVs controlled by a bigger mother AUV to balance cost effectiveness and functionality.
- However, data scarcity can make child AUV and target 3D pose estimation challenging.

Where Our Problem Lies.



Scarcity of Annotated Underwater Robot Images

Dataset	Images	Classes	Robots	Robots(%)
COU	9757	24	87	0.89
Seaclear	8610	40	58	0.67
DeepPlastic	3174	1	0	0.00
TrashCan	7212	22	2721*	37.73*
Trash-ICRA19	5700	9	1515*	26.58*
UNO	5902	22	<2721*	<46.10*

Method 1 (YOLOv9)

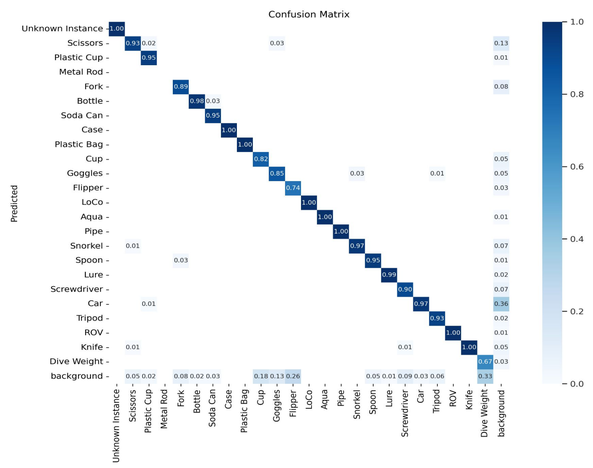
YOLOv9 Detection Pipeline

COCO pretraining
330k images,
80 classes
→
COU fine-tuning
6824 images,
24 classes

Model Performance On COU Test Set

Metric	Score
mAP@0.5	91.7%
mAP@0.5:0.95	85.1%
Precision	94.1%
Recall	85.1%

YOLOv9 Confusion Matrix



How can a mother AUV reliably estimate the relative positions of a child AUV and target objects in data-scarce underwater environments where annotated datasets are limited?

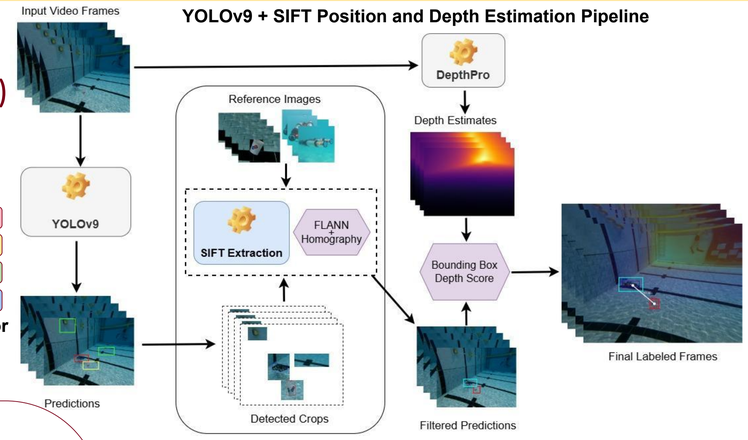
Method 2 (YOLOv9 + SIFT)

4
Target Reference Images

4
LoCO Reference Images

Front
Top
Back
Right

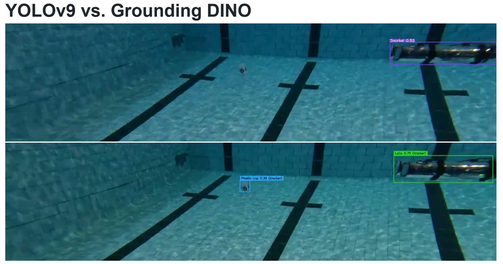
Number Of Images Used For SIFT Feature Matching.



Method 3
Grounding DINO + MIL tracking

- Combines image and text features to produce object detections.
- The pipeline runs the detector every 15 frames.
- Uses OpenCV Multiple Instance Learning (MIL) tracking between detection updates.

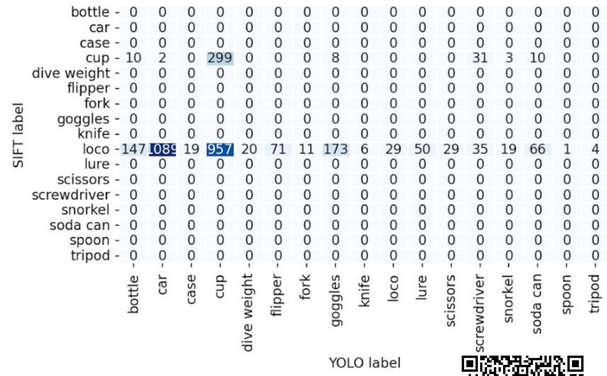
YOLOv9 vs. Grounding DINO



YOLOv9: LoCO receives the incorrect label “Snorkel” in this frame.

Grounding DINO + MIL: LoCO and the plastic cup receive the intended labels.

YOLOv9 + SIFT Corrected Label Counts



Scan to View References